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MACHINE LEARNING BASED FACTOR ESTIMATION IN FACTOR MODELS OF ASSET RETURNS

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PROCJENA FAKTORA U FAKTORSKIM MODELIMA POVRATA IMOVINA ZASNOVANA NA STROJNOM UČENJU

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Zadatak: **Procjena faktora u faktorskim modelima povrata imovina zasnovana na strojnom učenju**

Opis zadatka:

U ovom završnom radu razmatra se problem procjene faktora u faktorskim modelima financijskih povrata. U sklopu rada potrebno je prvo analizirati klasične pristupe procjeni faktora, uključujući metode zasnovane na glavnim komponentama i regresijskim postupcima. Nadalje, potrebno je razviti i implementirati pristupe zasnovane na metodama strojnog učenja za otkrivanje i procjenu latentnih faktora koristeći karakteristike imovina te odabrane makroekonomske varijable. Razvijene modele potrebno je empirijski ispitati na povijesnim tržišnim podacima, analizirati njihovu sposobnost objašnjavanja presjeka povrata te usporediti performanse s klasičnim faktorskim modelima. Posebnu pažnju potrebno je posvetiti interpretaciji procijenjenih faktora i njihovoj ulozi u objašnjavanju sistematskih izvora rizika.

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Contents

1 Introduction

Predicting the returns of financial assets is the first question that presents itself to most participants in financial markets. At first glance, returns behave like a random variable: the prices of stocks and other forms of assets change with no apparent regularity. Nevertheless, part of these movements can be explained by common sources of risk, which is precisely the fundamental idea behind factor models. They assume that the return of an individual asset is not an independent random variable, but a linear combination of its exposures to a small number of common risk factors that act on the market as a whole. Each such factor carries a risk premium, that is, the average reward an investor earns in the long run by taking on exposure to that risk. An asset that is more strongly exposed to a given risk should therefore on average deliver a higher expected return as well, so under this assumption the differences in average returns across assets reduce to differences in their exposures to systematic sources of risk.

Applying a factor model requires estimating two types of quantities. The first is the exposure of an individual asset to each of the factors, and the second is the risk premium that each factor carries, that is, the average return that exposure to that factor delivers. Estimating the premiums proves to be a considerably harder task. Since every factor is itself the return of a particular part of the market, estimating its premium amounts to predicting the future movement of that part of the market, which is in itself a very demanding task. It is often approached in a simple way: the premium is estimated as the historical average of the returns of a given factor. Such an estimate, however, is often unreliable. Financial data are extremely noisy, and premiums are not constant but change over time, so the historical average easily gives an unstable and biased picture of the future premium. A model built on such an estimate therefore has difficulty generalizing well, because an unstable premium poorly describes returns outside the period over which it was estimated.

Classical approaches to estimating factor models rely mainly on regression procedures. This thesis stays within that framework, but focuses on its most sensitive part, the estimation of risk premiums. It starts from the standard Fama-French model (Fama and French, 1993), and the fundamental question it addresses is whether machine learning methods can give a better estimate of the premiums than classical approaches. The central idea is to use a long short-term memory (LSTM) recurrent cell instead of a simple average. The architecture of such a model is well suited to working with time series, so the network can recognize patterns in the movement of the factors that the mean ignores. In addition to the historical values of the factors themselves, the network also receives selected macroeconomic variables as input, which can serve as predictors of the premiums and from which the model can learn dependencies that are not visible in the factor returns alone. The resulting estimates are then compared with the historical mean and with the Fama-MacBeth regression as classical approaches, and also with an ideal (*oracle*) estimate that can serve as a theoretical upper bound on performance.

In recent years, machine learning and deep learning have found wide and successful application in empirical asset pricing (Feng et al., 2023). These methods recognize complex and nonlinear patterns in large and noisy financial data that linear approaches often fail to explain, which makes them a natural candidate for the task of estimating factor premiums. The great expressiveness of these models, however, also carries the risk of *overfitting*. A sufficiently complex model, while fitting the historical data well, begins to describe the noise in them as well, so its accuracy on the data used in training says nothing about the true quality of the estimate. The success of a method is therefore judged by its performance on data that the model did not see during training, that is, by its ability to generalize.

The aim of this thesis is to apply the described methods for estimating factor premiums to real market data, to compare them with one another, and to assess how successfully they explain the differences in average returns across assets. The emphasis is on *out-of-sample* performance: on predicting returns from periods that were not available to the methods during estimation. The thesis is organized as follows. The second chapter presents the theoretical background of factor models and of the methods for estimating risk premiums, the third describes the data used and their processing, while the fourth presents and analyses the results obtained. Finally, the main conclusions are drawn and possible directions for further research are outlined.

2 Factor models of asset returns

2.1 Returns and excess returns over the risk-free rate

Ownership of a financial asset carries a right to future cash flows. A typical example is a stock: it represents a share of ownership in a company, and it brings its holder a part of the company's profit and the possibility of earning through a rise in its price, that is, in its value. Besides individual stocks, portfolios are also important, that is, sets of several stocks or of some other forms of assets.

The most important property of an asset is the return it brings to its holder. The return measures the relative change in the value of an asset over one time period, that is, how much the holder earned or lost relative to the amount invested. If we denote the price of asset i at the end of period t by $P_{i,t}$, and the price at the end of the previous period by $P_{i,t-1}$, we define the return of the asset as

$$R_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}}. \tag{2.1}$$

The numerator is the price change realized over the period, and dividing by the initial price expresses that change in relative terms, so the return does not depend on the absolute price level and is comparable across different assets. We follow the value of an asset at discrete points in time, where the length of the period between two consecutive points is arbitrary. In this thesis that period will be one month, so the index t denotes the ordinal number of the month observed.

In a real setting, capital is rarely allocated to a single asset. It is distributed among several of them. We call such a set of assets a *portfolio*. If we compose it of N assets, we assign to each a *weight* w_i that states what part of the total invested capital falls on asset

i. The weights are non-negative and their sum equals one, $\sum_{i=1}^{N} w_i = 1$. Since the return is the ratio of the change in value to the amount invested, the return of a portfolio equals the weighted average of the returns of its constituents (Cochrane, 2005, Ch. 1),

$$R_{p,t} = \sum_{i=1}^{N} w_i R_{i,t}. \tag{2.2}$$

A special role is played by an asset that is not exposed to market risk, so that its return is known in advance and does not depend on the movement of the rest of the market. We call that return the *risk-free rate of return* and denote it by $R_{f,t}$. As its measure one takes the yield on short-term government treasury bills, which are considered one of the safest investments.

If we subtract from the return of an asset the risk-free rate of the same period, we obtain a quantity that we call the *excess return over the risk-free rate* (Cochrane, 2005, Ch. 1). Since the entire thesis deals exclusively with that return, the notation $R_{i,t}$ will from here on stand for the expression $R_{i,t} - R_{f,t}$, that is, the excess return over the risk-free rate.

The risk-free rate represents the opportunity cost of investing: the yield we would earn even without exposing ourselves to any risk. Only the part of the return that exceeds it is the actual reward for the risk taken. A positive expected excess return says that exposure to risk pays off on average, and a negative one that the asset does not reward the risk taken.

It is precisely this part of the return that is at the centre of factor models. They do not seek to explain total returns, but that part of the return which represents the reward for exposure to systematic sources of risk. For this reason all quantities in the remainder of the thesis, unless indicated otherwise, refer to excess returns over the risk-free rate, and the expected excess return of an individual asset is denoted by $E(R_{i,t})$.

2.2 Definition and assumptions of the factor model

2.2.1 From correlated variables to factors

The factor model originates in multivariate statistics, in which one seeks to describe the correlation structure of a large number of variables by a small number of hidden, unobserved variables (Johnson and Wichern, 2013). It starts from a simple observation. If the variables can be grouped so that within a group they are strongly correlated with one another and weakly correlated with variables from other groups, it is natural to assume that behind each group stands one common, unobserved variable, a *factor*, responsible for that common movement. Factor analysis seeks to confirm the existence of such a structure and to replace the multitude of individual variables by factors.

The factors are most often latent and are estimated from the data. In this thesis, by contrast, we specify the factors in advance (Section 2.4.2). In the remainder of this section we first define the model and its assumptions, and then single out those that are relevant for the approach we use.

2.2.2 General definition of the model

We consider an observed random vector $X = (X_1, \dots, X_p)^\top$ with p components and expectation $\mu = E(X)$. The factor model assumes that X depends linearly on a small number $m < p$ of unobserved random quantities $F_1, \dots, F_m$, which we call *common factors*, and on p additional sources of variation $\varepsilon_1, \dots, \varepsilon_p$, which we call *errors* or *specific factors* (Johnson and Wichern, 2013). Component-wise the model reads

$$\begin{aligned} X_1 - \mu_1 &= \ell_{11}F_1 + \ell_{12}F_2 + \cdots + \ell_{1m}F_m + \varepsilon_1, \\ &\vdots \\ X_p - \mu_p &= \ell_{p1}F_1 + \ell_{p2}F_2 + \cdots + \ell_{pm}F_m + \varepsilon_p, \end{aligned} \tag{2.3}$$

or, in matrix notation,

$$\underset{(p\times1)}{X - \mu} = \underset{(p\times m)}{L}\,\underset{(m\times1)}{F} + \underset{(p\times1)}{\varepsilon} \tag{2.4}$$

(Johnson and Wichern, 2013, Eqs. 9-1, 9-2).

The coefficient ℓ_{ij} is called the *loading* of the i-th variable on the j-th factor, so $L =$

$[\ell_{ij}] \in \mathbb{R}^{p \times m}$ is the *matrix of factor loadings*. We denote its i-th row by $\beta_i^\top = (\ell_{i1}, \dots, \ell_{im})$, and it contains the exposures of the i-th variable to all factors. The specific factor ε_i is tied exclusively to the i-th variable.

Although the matrix form (2.4) resembles a linear regression model, there is an essential difference between them: in regression the variables are observed, whereas here both the loadings L and the factors F are unobserved. For this reason the model cannot be estimated by ordinary regression, but only under additional assumptions about F and ε, which we state below (Johnson and Wichern, 2013).

2.2.3 The systematic and the idiosyncratic part

We split the expression $X_i - \mu_i = \beta_i^\top F + \varepsilon_i$ into two parts. The first, $\beta_i^\top F$, is called the *systematic part*: it is made up of the same factors F for all variables, and it differs across variables only through the loadings β_i. This part carries the variation *common* to the whole set and is the source of the correlation among the variables. The second, ε_i, is called the *idiosyncratic* or *specific part*: it is tied exclusively to the i-th variable and carries the variation that the common factors do not explain.

The loading vector β_i measures the sensitivity of the i-th variable to the factors: a larger absolute value of ℓ_{ij} indicates a stronger exposure to factor j. The decomposition separates the systematic risk $\beta_i^\top F$, common to all variables, from the idiosyncratic risk ε_i, specific to an individual variable. Idiosyncratic risk is removed by diversification; this distinction is the basis of the pricing discussed in Section 2.3.

2.2.4 Assumptions of the model

For the decomposition into a common and a specific part to be meaningful, we impose the following assumptions on the factors and the errors (Johnson and Wichern, 2013):

- the model is linear in the factors, as written in (2.4);
- the errors are centred, $E(\varepsilon) = 0$;
- the factors and the errors are uncorrelated, $\mathrm{Cov}(F, \varepsilon) = 0$;
- the errors of different components are mutually uncorrelated, i.e. $\mathrm{Cov}(\varepsilon) = \Psi$ is a

diagonal matrix.

The last assumption is the core of the factor model: it says that all the correlation among the components of the vector X is carried exclusively by the common factors, while the specific errors share variation neither with one another nor with the factors.

2.3 Factor models in asset pricing

We now apply the general formulation of the previous section to asset returns. We no longer write the model as $X - \mu$, but as the excess return of an individual asset: in asset pricing the factors are observed and have expectations different from 0, and we interpret them as risk premiums. The notation now also contains an intercept α_i and the expectation of the factors $\lambda = E(F)$, which were not present in the initial definition.

The assumption that the idiosyncratic errors are uncorrelated, expressed by the diagonality of the matrix Ψ, has a key consequence for asset pricing. In a portfolio composed of a large number of assets the specific errors are mutually uncorrelated, so when summed they largely cancel one another out; the common exposure to the factors, by contrast, remains. An investor can therefore remove idiosyncratic risk almost entirely by diversifying investments, at no additional cost, so the market assigns it no reward (Cochrane, 2005, Ch. 1). Only systematic risk is rewarded, that is, exposure to the common factors.

Let us now consider the expected return of an individual asset. The factor model in excess returns over the risk-free rate reads

$$R_i = \alpha_i + \beta_i^\top F + \varepsilon_i, \tag{2.5}$$

where $\beta_i^\top F$ is the systematic part and ε_i the idiosyncratic error with zero expectation. Taking the expectation of both sides of Equation (2.5), the idiosyncratic part vanishes and there remains

$$E(R_i) = \alpha_i + \beta_i^\top \lambda, \qquad \lambda = E(F). \tag{2.6}$$

The vector λ contains the expectations of the factors, and we interpret each of its components as a *risk premium*, the average reward for a unit of exposure to the corresponding factor. The product $\beta_i^\top \lambda$ is therefore that part of the expected return which stems from the asset's exposure to systematic risks.

The remaining term α_i is the part of the expected return that can *not* be attributed to exposure to the factors, the so-called *pricing error*. If the factor model fully explains the return of an asset, this term vanishes. Under the restriction $\alpha_i = 0$ expression (2.6) reduces to

$$E(R_i) = \beta_i^\top \lambda, \tag{2.7}$$

so the differences in expected returns across assets are fully determined by the differences in their exposures β_i and by the common premiums λ.

Equation (2.7) summarizes the goal of asset pricing with a factor model: to explain expected returns by means of two types of quantities, the exposures β_i specific to each asset and the premiums λ common to all assets. Estimating the premiums λ is the more delicate of the two and, as pointed out in the introduction, is the central problem addressed in this thesis. What follows is devoted to the estimation of λ.

2.4 Examples of factor models: CAPM and Fama-French

We have established that if returns are driven by a small number of common sources of risk, expected returns depend on the exposures to those factors and on their premiums. The key empirical question remains: which factors to choose. Historically, the answer developed from the simplest model with a single, market factor towards models with several factors. Below we first define the CAPM, and then the Fama-French three-factor model that we use in the rest of the thesis.

2.4.1 Capital Asset Pricing Model (CAPM)

The simplest and historically first model of this type is the Capital Asset Pricing Model (CAPM) (Sharpe, 1964). It assumes the existence of only one common source of risk, the market as a whole, so it is a factor model with $m = 1$. The corresponding factor is the *excess market return over the risk-free rate*, $\mathrm{MKT} = R_M - R_f$, where R_M is the return of the *market portfolio*, a portfolio that comprises all risky assets with a weight that depends on market capitalization. In reality that portfolio is approximated by one of the broad market indices. The market factor thereby represents the common movement of the entire market, that is, the systematic risk to which all assets are exposed, to differing degrees.

Substituting this single factor into the general expression (2.7), the expected return of an asset becomes proportional to its market exposure,

$$E(R_i) = \beta_i^{\mathrm{MKT}} \lambda_{\mathrm{MKT}}, \tag{2.8}$$

where β_i^{MKT} is the asset's sensitivity to the market factor and λ_{MKT} is the market risk premium. The CAPM thereby claims that all differences in expected returns across assets reduce to differences in a single quantity, the market exposure β_i^{MKT}.

2.4.2 The Fama-French three-factor model

Although conceptually sound and powerful, the CAPM does not manage to explain the entire set of average returns. It turned out that average returns also depend on characteristics that the market factor does not capture: companies with a small market capitalization and companies with a high book-to-market ratio deliver on average higher returns than the CAPM would predict (Fama and French, 1993). In order to capture these regularities, Fama and French (Fama and French, 1993) extend the model with two additional factors.

Alongside the market factor MKT, the model introduces:

- the SMB factor (*small minus big*), the difference between the average returns of portfolios of companies with small and large market capitalization;
- the HML factor (*high minus low*), the difference between the average returns of portfolios of companies with a high and a low book-to-market ratio.

The factors SMB and HML are therefore returns of zero-investment portfolios (a long position in one group of companies, a short position in the other), while the market factor is already expressed as an excess return over the risk-free rate. The three-factor model for asset i then reads

$$R_i = \alpha_i + \beta_i^{\mathrm{MKT}}\,\mathrm{MKT} + \beta_i^{\mathrm{SMB}}\,\mathrm{SMB} + \beta_i^{\mathrm{HML}}\,\mathrm{HML} + \varepsilon_i, \tag{2.9}$$

which is a special case of the general model (2.5) with $F = (\mathrm{MKT}, \mathrm{SMB}, \mathrm{HML})^\top$.

All three factors are returns of concrete portfolios, which is an important property of theirs. As shown in Section 2.3, the risk premium of a factor equals its expectation, $\lambda_k = E(F_k)$. Since each factor is itself a return, its premium reduces to the expected return of the corresponding portfolio, so estimating the premiums λ also reduces to estimating the expected returns of the three factors. The rest of the thesis is devoted to that problem.

We emphasize that the Fama-French model is only *one* of the possible choices of factors; here we take it as a standard framework within which we study the estimation of the premiums, and not as an object of analysis in its own right. The details of the construction of the factors and the source of the data are defined in Chapter 3.

2.5 Estimation of the factor model

So far we have only defined the factor model and its special cases. In order to apply it to real data, we must estimate its parameters. We distinguish two types of quantities: the factor exposures β_i, specific to each asset, and the risk premiums λ, common to all assets. Both are estimated from observed returns, and the basic method for doing so is linear regression by the method of least squares.

2.5.1 Linear regression and the method of least squares

Linear regression describes the relationship of one dependent variable with one or more explanatory variables. Let y be the dependent variable and $x = (x_1, \dots, x_k)^\top$ a vector of k explanatory variables; y is assumed to be approximately a linear function of x,

$$y_t = b_0 + b_1 x_{t1} + \cdots + b_k x_{tk} + u_t, \qquad t = 1, \dots, n, \tag{2.10}$$

where $b_0, \dots, b_k$ are unknown coefficients, u_t is the error (the deviation of an observation from the linear relationship), and n is the number of observations. Collecting the observations of the dependent variable into a vector $y \in \mathbb{R}^n$, and the regressors into a *design matrix* $\mathbf{X} \in \mathbb{R}^{n\times(k+1)}$ whose first column consists of ones (for the intercept b_0), the model reads compactly

$$y = \mathbf{X}b + u, \qquad b = (b_0, b_1, \dots, b_k)^\top. \tag{2.11}$$

We estimate the coefficients by the *method of least squares* (*ordinary least squares*, OLS) (Hastie et al., 2009, Ch. 3): we choose those values that minimize the sum of squared residuals, that is, the sum of squared deviations of the observed values from those that the model predicts,

$$\hat{b} = \arg\min_b \sum_{t=1}^{n} \left(y_t - b_0 - b_1 x_{t1} - \cdots - b_k x_{tk}\right)^2. \tag{2.12}$$

Under the assumption that the matrix $\mathbf{X}$ has full rank, this minimization problem has a unique closed-form solution,

$$\hat{b} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top y. \tag{2.13}$$

Geometrically, $\hat{b}$ determines the linear function closest to the observed data in the sense of the smallest sum of squared deviations. We use this same procedure twice: in Section 2.5.2 to estimate the factor exposures β_i (by regressing returns on the factors), and within the Fama-MacBeth method to estimate the premiums λ (by a cross-sectional regression).

2.5.2 Estimation of the factor exposures

In the abstract setting of the factor model the factors are unobserved: their values cannot be measured directly, but must be estimated from the data together with the loadings. In the Fama-French model the situation is somewhat different. The factors are given as returns of concrete portfolios, so their values in every period are directly available from market data. The factor exposures β_i can therefore be estimated directly by a linear regression of the asset's excess returns over the risk-free rate on the factors.

Consider an asset i and the series of its excess returns over the risk-free rate $R_{i,t}$ over T consecutive periods. According to model (2.5), every excess return is a linear function of the values of the factors at time t,

$$R_{i,t} = \alpha_i + \beta_i^\top F_t + \varepsilon_{i,t}, \qquad t = 1, \dots, T, \tag{2.14}$$

where $F_t = (\mathrm{MKT}_t, \mathrm{SMB}_t, \mathrm{HML}_t)^\top$ is the vector of factors in period t. This is precisely the linear regression model (2.11): the dependent variable is the asset's excess return over the risk-free rate, the regressors are the factors, the intercept is α_i, and the slopes are the sought exposures β_i. Applying the method of least squares (2.13) yields the estimates $\hat{\alpha}_i$ and $\hat{\beta}_i$. Such a regression, in which the return of an individual asset over time is regressed on the factors, we call a *time-series regression*.

The time-series regression gives the exposure estimates $\hat{\beta}_i$ for all assets. This leaves only the second group of quantities unestimated: the risk premiums λ, to whose estimation methods the next section is devoted.

2.6 Methods for estimating the risk premiums

Since for the Fama-French factors the premiums equal the expectations of the factors, $\lambda_k = E(F_k)$ (Section 2.3), estimating λ reduces to estimating the expected returns of the factors. In this section we define several approaches to that estimation. Each of them gives its own estimate $\hat{\lambda}$, from which the predicted expected excess return of an asset over the risk-free rate follows. We denote that estimate of the expectation by $\hat{\mu}_i$,

$$\hat{\mu}_i = \hat{E}(R_i) = \hat{\beta}_i^\top \hat{\lambda}. \tag{2.15}$$

2.6.1 Estimation by the historical mean

The simplest way to estimate the expectation of any quantity is its average in the sample. Since the premium of a factor is precisely its expectation, it is natural to estimate it by the average of the observed historical factor returns,

$$\hat{\lambda}^{\text{hist}} = \frac{1}{T} \sum_{t=1}^{T} F_t, \tag{2.16}$$

where $F_1, \dots, F_T$ are the factor returns over the T periods available at the time of estimation. The premium of each factor is thereby estimated by its historical average return.

The approach is intuitive and introduces no additional assumptions, so it serves as the baseline against which we compare the other methods. Its weakness is that it assumes the premium is constant over time. Since financial returns are very noisy, and premiums change over time, the historical average can give an unstable and biased estimate, which is precisely what motivates the more complex approaches that follow.

2.6.2 Estimation with known future factors (*oracle*)

The historical mean uses only data known at the time of estimation. In order to separate the error that stems from the *estimation of the premiums* from the error that stems from the model itself, we introduce an idealized (*oracle*) estimate that deliberately uses information unavailable in reality.

Instead of historical returns, the oracle estimates the premiums by the average of the factor returns in exactly that future period for which the prediction is made,

$$\hat{\lambda}^{\text{ora}} = \frac{1}{H} \sum_{h=1}^{H} F_{t+h}, \tag{2.17}$$

where $F_{t+1}, \dots, F_{t+H}$ are the factor returns in the prediction period of length H. In other words, the oracle estimates the premiums *by looking into the future*, as if at the time of estimation it already knew the future factor returns perfectly. For this reason it is not a feasible prediction method, but a perfectly informed idealization that serves us solely as a benchmark.

Its purpose is to serve as a reference. The oracle gives the best estimate that any other method for estimating λ could achieve given the exposures $\hat{\beta}_i$, because it removes every error in the estimation of the premiums. The remaining prediction error then comes solely from the limitations of the model: imperfect exposures and idiosyncratic noise. The oracle thereby serves as an *upper bound on performance*, and the difference between the performance of a feasible method and that of the oracle measures what it actually costs us to have to estimate the premiums.

2.6.3 The Fama-MacBeth regression

The historical mean and the oracle estimate the premiums directly from the returns of the factors themselves. The Fama-MacBeth regression (Fama and MacBeth, 1973) takes a different route: it estimates the premiums from the *cross-section of asset returns*. In each period it examines how large a reward the market actually paid in that month per unit of exposure to each factor, and then averages those rewards over time. The procedure is carried out in two steps.

First step (time-series regression). For each asset we estimate its factor exposures $\hat{\beta}_i$ by a time-series regression of the excess returns over the risk-free rate on the factors, exactly by the procedure of Section 2.5.2. The result is the estimated exposures of all assets.

Second step (cross-sectional regression). In each period t we regress the *cross-section* of the excess returns over the risk-free rate of all N assets on their estimated exposures,

$$R_{i,t} = \gamma_{t,0} + \gamma_{t,1}\hat{\beta}_{i,1} + \cdots + \gamma_{t,m}\hat{\beta}_{i,m} + e_{i,t}, \qquad i = 1, \dots, N. \tag{2.18}$$

Now the exposures $\hat{\beta}_i$ are the known regressors, and the unknowns are the coefficients $\gamma_{t,k}$. The slope $\gamma_{t,k}$ measures the additional return that in period t accrued to one unit of higher exposure to factor k, that is, the return (premium) of factor k realized in that period; the intercept $\gamma_{t,0}$ is the part of the return unrelated to the exposures. By repeating this cross-sectional regression in every period we obtain a time series of estimated factor returns $\gamma_t = (\gamma_{t,1}, \dots, \gamma_{t,m})$.

Estimation of the premiums. Finally, we estimate the premium of each factor by the average of its returns over all T periods,

$$\hat{\lambda}^{\mathrm{FM}} = \bar{\gamma} = \frac{1}{T}\sum_{t=1}^{T}\gamma_t, \qquad \bar{\gamma}_k = \frac{1}{T}\sum_{t=1}^{T}\gamma_{t,k}. \tag{2.19}$$

Both regressions, the time-series and the cross-sectional one, are carried out by the method of least squares of Section 2.5.1. Unlike the historical mean, which uses only the factor returns, Fama-MacBeth extracts the premiums from the relationship between exposures and returns across the entire set of assets. If the model explains average returns well, the average intercept $\bar{\gamma}_0$ should thereby be close to zero, in accordance with the restriction $\alpha_i = 0$ of Section 2.3.

2.7 Estimation of the premiums by deep learning

The methods of the previous section estimate the premiums by a simple average or by regression. Let us now introduce an approach based on deep learning, which seeks to *predict* the premiums from the recent history of the factors by means of a recurrent neural network. We first briefly describe the network itself, and then the way in which we use it to estimate the premiums.

2.7.1 Recurrent networks and the LSTM

A neural network is a parameterized function that computes an output from an input and learns from data by adjusting its parameters. The data in this thesis are a time series, a sequence of factor returns, in which the ordering carries information, so a network is needed that processes this sequence while taking the past into account. A *recurrent neural network* (RNN) has that property. It processes the sequence step by step and at every time t maintains a *hidden state* $h^{(t)}$ that summarizes what has been seen so far,

$$h^{(t)} = f\big(h^{(t-1)}, x^{(t)}\big)\,, \tag{2.20}$$

where $x^{(t)}$ is the input at time t. The hidden state is carried from step to step, so in later steps the network can use information from earlier ones.

The *long short-term memory* (LSTM) recurrent cell solves the difficulties that arise in learning over long sequences by separating the *cell state* $c^{(t)}$, which serves as memory, from the hidden state $h^{(t)}$ from which the output is computed (Goodfellow et al., 2016, Ch. 10.10). The cell state is updated by addition, without multiplication by a matrix, which avoids those difficulties:

$$c^{(t)} = f^{(t)} \odot c^{(t-1)} + i^{(t)} \odot \hat{c}^{(t)}, \tag{2.21}$$

where $\hat{c}^{(t)}$ is the *candidate contribution to the cell state*, computed from the current input and the previous hidden state, and $\odot$ denotes element-wise multiplication (the Hadamard product).

The quantities $f^{(t)}$ and $i^{(t)}$ are *gates* that filter information: each is a vector of values in the interval $(0, 1)$, obtained by a sigmoid function, which determines how much of the information is let through. The *forget gate* $f^{(t)}$ determines how much of the previous cell state is retained, and the *input gate* $i^{(t)}$ how much of the new contribution $\hat{c}^{(t)}$ is added. Finally, we compute the hidden state from the cell state by means of the *output gate* $o^{(t)}$,

$$h^{(t)} = o^{(t)} \odot \tanh\left(c^{(t)}\right). \tag{2.22}$$

By separating the memory $c^{(t)}$ from the output $h^{(t)}$, by the additive update of the cell state and by the three gates that filter information, the LSTM retains important information across longer sequences and avoids vanishing and exploding gradients. It is precisely this ability to work with long time series that makes it suitable for estimating factor premiums from their history, which we turn to next.

2.7.2 Estimation of the premiums by an LSTM network

The last method is based on the LSTM network of the previous section. Instead of estimating the premiums by the average of past returns, we train the network to predict them from the recent history of the factors: as input we give it a sequence of past factor values, and as output it gives an estimate of the premiums $\hat{\lambda}^{\mathrm{LSTM}}$, from which the predicted expected excess return over the risk-free rate follows as before, $\hat{\mu}_i = \hat{\beta}_i^{\top} \hat{\lambda}^{\mathrm{LSTM}}$.

We train the parameters of the network on historical data, so that its predictions match the actually realized factor returns as well as possible. Unlike the historical mean, which assigns the same fixed value to all periods, the LSTM can account for the time variation and the patterns in the movement of the factors that a simple average ignores. The concrete form of the input and the training procedure are defined together with the experimental design later in the thesis.

3 Data

In order to apply the premium estimation methods of the previous chapter to real data and to compare them with one another, historical market data are needed. In this chapter we describe the data used: the assets whose returns we seek to explain, the factors, the risk-free rate, and the additional macroeconomic variable that we use later. For each source we state where it comes from and which period it covers, and we describe the procedure by which the individual datasets were prepared and merged into a single dataset for analysis.

3.1 Sources and sample period

The data were taken from two publicly available sources. The first is the data library of Kenneth French (French, 2024), from which we take the returns of the assets whose behaviour we explain, the values of the factors, and the risk-free rate. The second is the publicly available database of Robert Shiller, from which we take the macroeconomic variable described later in this chapter.

All quantities are observed at the *monthly* level, so one period corresponds to one month, in accordance with the notation introduced in Section 2.1. Returns are expressed in percent. The sample covers the period from January 1970 to February 2026, a total of 674 months. We choose 1970 as the beginning because from then on all three groups of data (asset returns, factors and the macroeconomic variable) exist simultaneously and have no gaps, so that after merging them we obtain an uninterrupted time series with no missing values. The estimates by each of the defined methods are carried out on the dataset obtained in this way.

3.2 Test assets

We apply the model to a set of assets whose returns we want to explain, which we call the test assets. On them we later assess how successful the individual methods for estimating the premiums are. In this thesis these assets are the 49 industry portfolios taken from Kenneth French's data library.

The industry portfolios are formed by classifying every company on the U.S. market, on the basis of its main line of business, into one of 49 industries (for example banking, mining or food production). All companies of the same industry then form one portfolio, whose monthly return we observe. These are portfolios in which the weights are determined by market capitalization. Within each portfolio an individual company participates in proportion to its market size, so that larger companies have a greater weight. This is precisely the weighted average of returns introduced in Section 2.1, except that the weights here are not equal, but reflect market capitalization.

In this way we obtain 49 time series of monthly returns, one for each industry, which we follow throughout the sample period. In accordance with what was defined in Section 2.1, here too we work with excess returns over the risk-free rate, so from the return of each portfolio we subtract the risk-free rate of the same month.

We choose the industry portfolios as test assets because they provide a broad cross-section of the market: they cover different economic sectors, which differ from one another in their sensitivity to the factors, and so give diverse exposures on which the model can show its worth. It is equally important that they are not formed according to the same characteristics on which the factors themselves rest (size and the book-to-market ratio), so that the estimation is not adapted to the factors in advance and the performance of the factors is tested more thoroughly. The diversity of average returns across industries is shown in Figure 3.1: the average monthly excess return over the risk-free rate ranges from 0.24 % to 1.06 %, which confirms that different industries have appreciably different average returns.

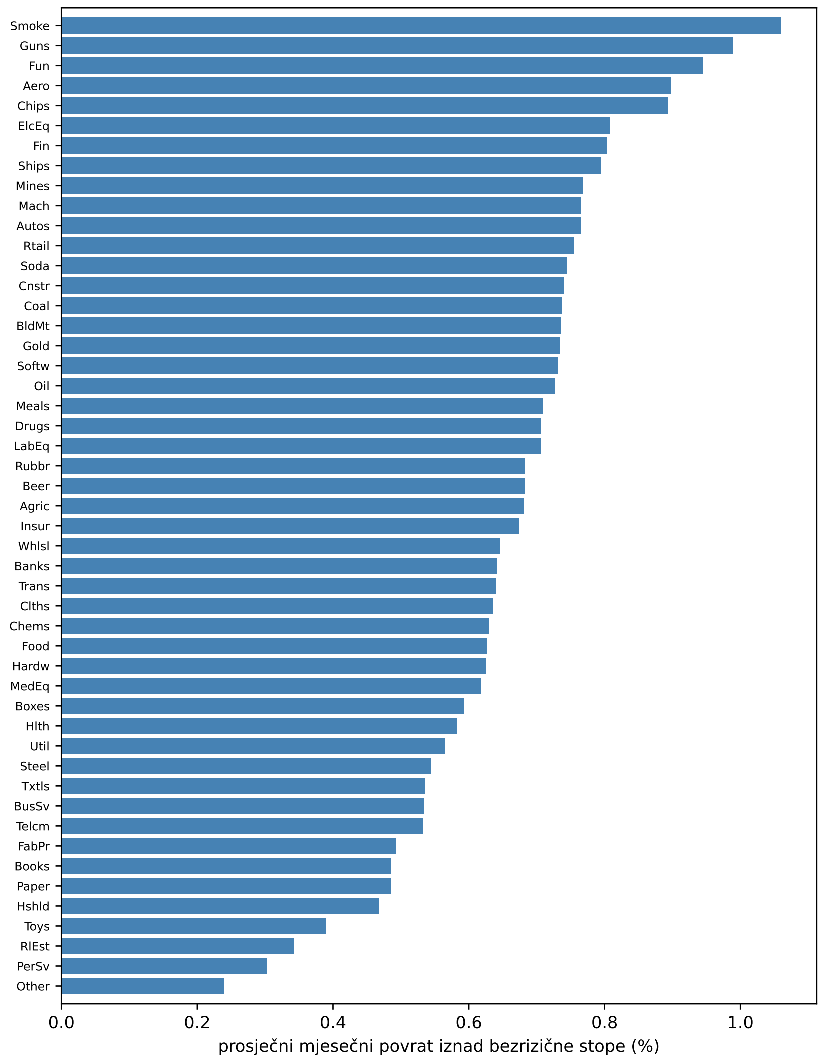

Figure 3.1: Average monthly excess return over the risk-free rate for each of the 49 industries, sorted by magnitude.

3.3 The factors and the risk-free rate

In this thesis we use the three Fama-French factors (the market factor MKT, SMB and HML) introduced in Section 2.4.2. All three factors, as well as the risk-free rate, are taken from Kenneth French's data library. They are expressed in percent at the monthly level, just like the returns of the test assets.

The market factor MKT is the excess return over the risk-free rate of a very broad market portfolio, that is, the market-capitalization-weighted return of all the companies covered, less the risk-free rate. The risk-free rate R_f, introduced in Section 2.1, is here specifically the yield on one-month U.S. government treasury bills, taken from the same source.

The factors SMB and HML are built from the returns of six auxiliary portfolios obtained by a double sort of companies (Fama and French, 1993). The companies are first divided by market capitalization into two groups, small (S) and big (B), and then by the book-to-market ratio into three groups: low (L), medium (M) and high (H). Applying these divisions produces six portfolios, which we label according to their size group and their ratio group, for example $R_{S,H}$ for the return of the portfolio of small companies with a high ratio. The SMB factor is defined as the difference between the average return of the three portfolios of small companies and the average return of the three portfolios of big companies,

$$\mathrm{SMB} = \tfrac{1}{3}\big(R_{S,L} + R_{S,M} + R_{S,H}\big) - \tfrac{1}{3}\big(R_{B,L} + R_{B,M} + R_{B,H}\big), \tag{3.1}$$

while the HML factor is the difference between the average return of the two portfolios with a high ratio and the average return of the two portfolios with a low ratio,

$$\mathrm{HML} = \tfrac{1}{2}\big(R_{S,H} + R_{B,H}\big) - \tfrac{1}{2}\big(R_{S,L} + R_{B,L}\big). \tag{3.2}$$

In this way both differences, in accordance with Section 2.4.2, isolate the desired effect: SMB the difference by size, and HML the difference by the book-to-market ratio.

Basic descriptive statistics of the three factors over the sample period are shown in Table 3.1. The market factor has the largest average value, but also the largest variability, while the SMB and HML factors are on average smaller. The correlations among the factors are small, which is desirable in this case, because it shows that each factor is a separate source of distinct information.

Table 3.1: Descriptive statistics of the monthly factors over the sample period: mean and standard deviation (in percent), together with their mutual correlations.

			Correlations		
Factor	Mean (%)	Std. dev. (%)	MKT	SMB	HML
MKT	0.618	4.569	1.00	0.28	−0.22
SMB	0.080	3.041	0.28	1.00	−0.15
HML	0.305	3.073	−0.22	−0.15	1.00

In order to show the long-run behaviour of the three factors over the observed period, we plot their cumulative movement in Figure 3.2.

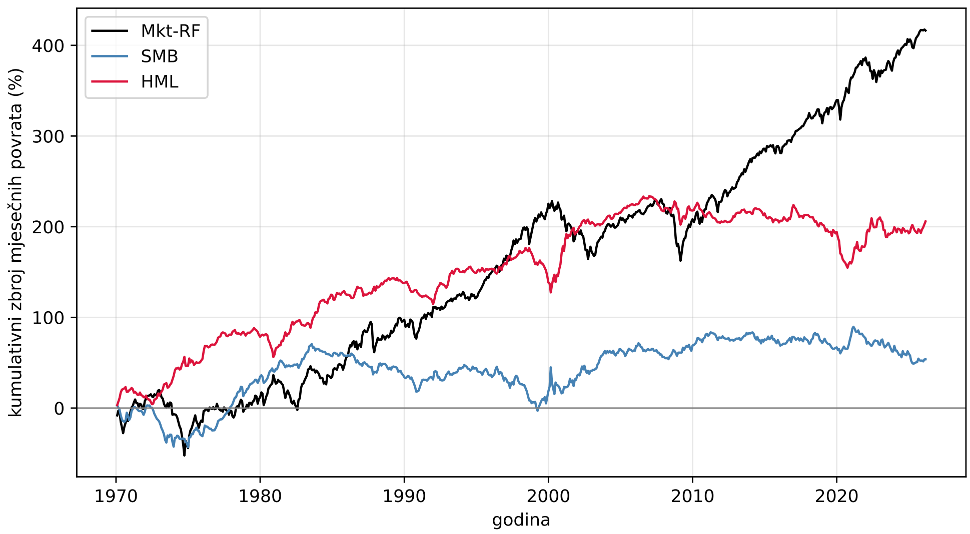

Figure 3.2: Cumulative sum of the monthly returns of the three factors over the sample period. Each curve shows the accumulated returns of one factor from the beginning of the period up to time t. It shows the level of the premium and its variability over time.

3.4 Shiller's price-to-earnings ratio

Alongside the factors and the risk-free rate, we also introduce into the analysis a macroeconomic variable that describes an adjusted relationship between a company's market capitalization and its net earnings. This is the *excess CAPE yield* (ECY), a quantity derived from Shiller's CAPE indicator, which we take from Shiller's online database (Shiller, 2024).

CAPE (the *cyclically adjusted price-to-earnings ratio*) is the price-to-earnings ratio adjusted for the business cycle. Unlike the ordinary price-to-earnings ratio, which has the earnings of a single year in the denominator, CAPE takes in the denominator the average of real earnings over the past ten years (Shiller, 2024). Formally, with the real price P and the real earnings E_j realized in year j of the past ten, CAPE is

$$\text{CAPE} = \frac{P}{\frac{1}{10}\sum_{j=1}^{10} E_j}. \tag{3.3}$$

By averaging earnings over a ten-year window, the influence of short-term variations connected with the business cycle is removed, so that CAPE gives a more stable measure of the relative value of the market with respect to its profitability. Its reciprocal value, $1/\text{CAPE}$, represents the cyclically adjusted *earnings yield*, that is, an estimate of the yield that stocks deliver relative to the price paid.

The excess CAPE yield is then the difference between the yield just mentioned and the real (inflation-adjusted) long-term interest rate. In this way it measures how much more is expected from investing in stocks than from relatively safe long-term bonds, so it serves as a valuation-based indicator of the expected return of stocks above the risk-free rate. We emphasize that the interest rate that enters this calculation is a long-term real rate, different from the short-term risk-free rate R_f of Section 2.1. Both refer to safe yields, but they differ in maturity and in their role in the model.

Unlike the factors, this variable does not enter the factor model itself; instead we use it as an additional input in the estimation of the premiums by deep learning (Section 2.7). In this way, alongside the history of the factors themselves, we also give the network broader information about the current over- or undervaluation of the market, while the classical estimation methods do not use this variable. The value is available at the monthly level throughout the sample period. Its movement, together with the risk-free rate, is shown in Figure 3.3. It is evident that the excess CAPE yield varies appreciably over time, unlike the small and almost constant risk-free rate, so we can say that it carries information about the changing level of market valuation that the risk-free rate alone does not provide.

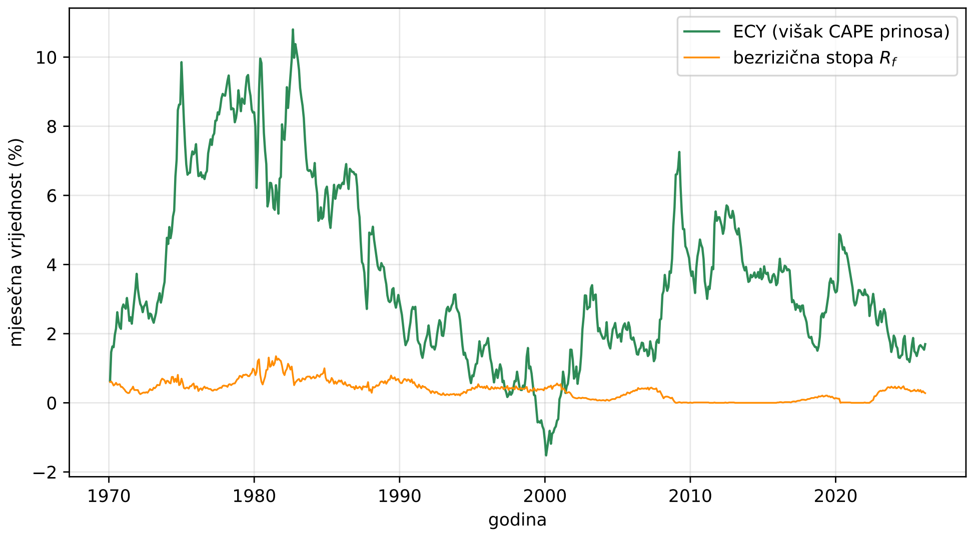

Figure 3.3: The excess CAPE yield (ECY) and the risk-free rate R_f over the sample period, both at the monthly level in percent.

3.5 Processing

Before the analysis we bring the initial data from the individual sources into a usable form. Each series is reduced to a monthly value dated at the end of the month, and values that indicate missing data are marked separately. All sources are then restricted to a common time range: we keep only the periods from January 1970 onwards, in which asset returns, the factors, the risk-free rate and the macroeconomic variable exist simultaneously. This gives us a single dataset without gaps.

From the return of each portfolio we subtract the risk-free rate of the same month in order to obtain excess returns over the risk-free rate, in accordance with Section 2.1. The excess CAPE yield is thereby expressed in percent, on the same scale as the other returns.

4 Results

In this chapter we apply the premium estimation methods of the previous chapters to the data described and compare them with one another. We first set out the procedure by which we carry out the methods, then the measures by which we assess performance, and after that the results of the three classical methods and of the method based on deep learning, which we finally compare with one another.

4.1 Rolling time window

The goal is not to explain past returns, but to predict future ones: a method that describes well precisely the period on which it was trained, but predicts poorly outside it, is not usable. We therefore separate estimation and evaluation in time, by a procedure that we call a *rolling time window*.

The rolling window works in the following way. At each step we consider 60 consecutive months, which we call the *estimation period*, and the 12 months that immediately follow and form the *prediction period*. On the estimation period we estimate the parameters of the model, and on the prediction period, which was not used in the estimation, we check how accurate the predictions are. We then shift the whole window forward by one month and repeat the procedure, until we exhaust the sample. This produces a large number of predictions, each for a different period, on the basis of which we can compare the methods reliably. Since the prediction period always lies after the estimation period, this is an *out-of-sample* evaluation.

Within each step we proceed identically for all methods. First, for each of the 49 industries we estimate its factor exposures, by a time-series regression of its excess returns over the risk-free rate on the factors over the 60 months of the estimation period, as described in Section 2.5.2. We then estimate the risk premiums by one of the methods of Sections 2.6 and 2.7. The predicted excess return over the risk-free rate for industry i is then

$$\hat{\mu}_i = \hat{\beta}_i^\top \hat{\lambda}, \tag{4.1}$$

the product of the estimated exposures and the estimated premiums. The prediction does not contain the intercept α_i: according to what was defined in Section 2.3, the expected excess return over the risk-free rate reduces entirely to the exposure to the factors and their premiums, and therefore in the prediction we take $\alpha_i = 0$.

As the outcome we take the average excess return over the risk-free rate of that industry over the 12 months of the prediction period. We call that quantity the realized return, and we compare the predictions with it. The difference between the predicted and the realized return, collected across all industries and all window positions, is the basis for the performance measures that we introduce below.

4.2 Performance measures

We assess the predictions according to how close they are to the realized returns. Let us denote by $r_{i,w}$ the realized and by $\hat{r}_{i,w}$ the predicted excess return over the risk-free rate of industry i in window w, and let N be the total number of such returns, including all industries and all positions of the rolling window. We measure performance in two ways.

The first measure is the *mean squared error* (MSE), the average of the squared deviations of the prediction from the realized return,

$$\text{MSE} = \frac{1}{N} \sum_{i,w} \left(r_{i,w} - \hat{r}_{i,w} \right)^2. \tag{4.2}$$

Through squaring, all errors become positive, and larger ones are penalized more heavily. A smaller value of the MSE therefore comes from better predictions.

The second measure is the *coefficient of determination* R^2, which relates the performance of a method to a simple null model. The null model is a prediction that would, for each industry, simply use that industry's own average realized return $\bar{r}_i$ over the entire period. The coefficient of determination is then

$$R^2 = 1 - \frac{\sum_{i,w}\left(r_{i,w} - \hat{r}_{i,w}\right)^2}{\sum_{i,w}\left(r_{i,w} - \bar{r}_i\right)^2}. \tag{4.3}$$

In the numerator is the sum of squared errors of the method, and in the denominator the sum of squared deviations of the realized returns from that same industry average. If the errors of the method equal those in the denominator, $R^2 = 0$: the method is no better than the simple prediction by each industry's own average. A positive R^2 says that the method surpasses that benchmark, and the closer it is to the value 1, the more accurate the predictions. Since we compare each industry with its own average level of returns, the differences in average returns across industries do not affect the result.

4.3 Results of the classical estimation methods

In the rolling window of Section 4.1 we first apply the three classical methods for estimating the premiums described in Section 2.6: estimation by the historical mean, the oracle, and the Fama-MacBeth regression. We recall that the oracle method estimates the premiums from the average of the factors over the prediction period itself, and that it looks into the future and therefore cannot be used in practice. We take it as an idealized upper bound on performance. The evaluation is carried out on the entire set of out-of-sample predictions, which comprises 602 window positions and a total of 29,498 predictions.

The coefficient of determination R^2 and the MSE of each method are shown in Table 4.1. The results are, as expected, poor: both feasible methods, the historical mean and the Fama-MacBeth regression, have a negative R^2, which means that they predict worse than the null model, each industry's own average. Only the oracle achieves an appreciably positive R^2 and the smallest MSE. In other words, neither the historical mean nor the Fama-MacBeth regression manages to guess the premiums in advance better than a simple average, while only an estimate that makes use of future data succeeds in doing so. The results obtained confirm how difficult a problem the estimation of the premiums is, which is also the starting motivation of this thesis.

Table 4.1: Coefficient of determination R^2 and MSE of the three classical methods on the entire set of out-of-sample predictions (602 windows, 29,498 predictions).

Method	R^2	MSE
historical mean	−0.229	4.386
oracle	0.455	1.947
Fama-MacBeth	−0.337	4.773

The weakness of the estimates is also visible in Figure 4.1, which for each of the three factors compares the estimates of the historical mean and of the Fama-MacBeth regression with the oracle value. If the estimates were accurate, the points would lie on the dashed line of equality. Instead, they are widely scattered, which shows that the estimated premiums follow the actually realized values poorly.

Figure 4.2 shows R^2 separately for each of the 49 industries. The pattern is clear: the oracle achieves a positive R^2 in almost all industries, while the historical mean and the Fama-MacBeth regression are negative almost everywhere. The difference between the two feasible methods is in favour of the historical mean, which in most industries is better than the Fama-MacBeth regression.

The question remains whether a method based on deep learning can surpass these classical approaches and come closer to the upper bound set by the oracle, which is what we address next.

4.4 Estimation by deep learning

We build the last method on the LSTM network of Section 2.7. Unlike the three classical methods, which obtain the premiums by a direct computation from the window, here we *train* the network to predict the premiums from the recent history.

The input of the network is a sequence of 60 months, the same length as the estimation period, and in each month it contains five values: the three factors and two additional variables, the macroeconomic variable ECY and the risk-free rate (Sections 3.3 and 3.4). The output is the three estimated premiums $\hat{\lambda}^{\text{LSTM}} = (\hat{\lambda}_{\text{MKT}}, \hat{\lambda}_{\text{SMB}}, \hat{\lambda}_{\text{HML}})$, from which the predicted excess return over the risk-free rate follows as with the other methods, $\hat{\mu}_i = \hat{\beta}_i^{\top} \hat{\lambda}^{\text{LSTM}}$, according to Equation (4.1). The network itself consists of one LSTM layer with sixteen units and built-in *dropout* (rate 0.3), which during training randomly

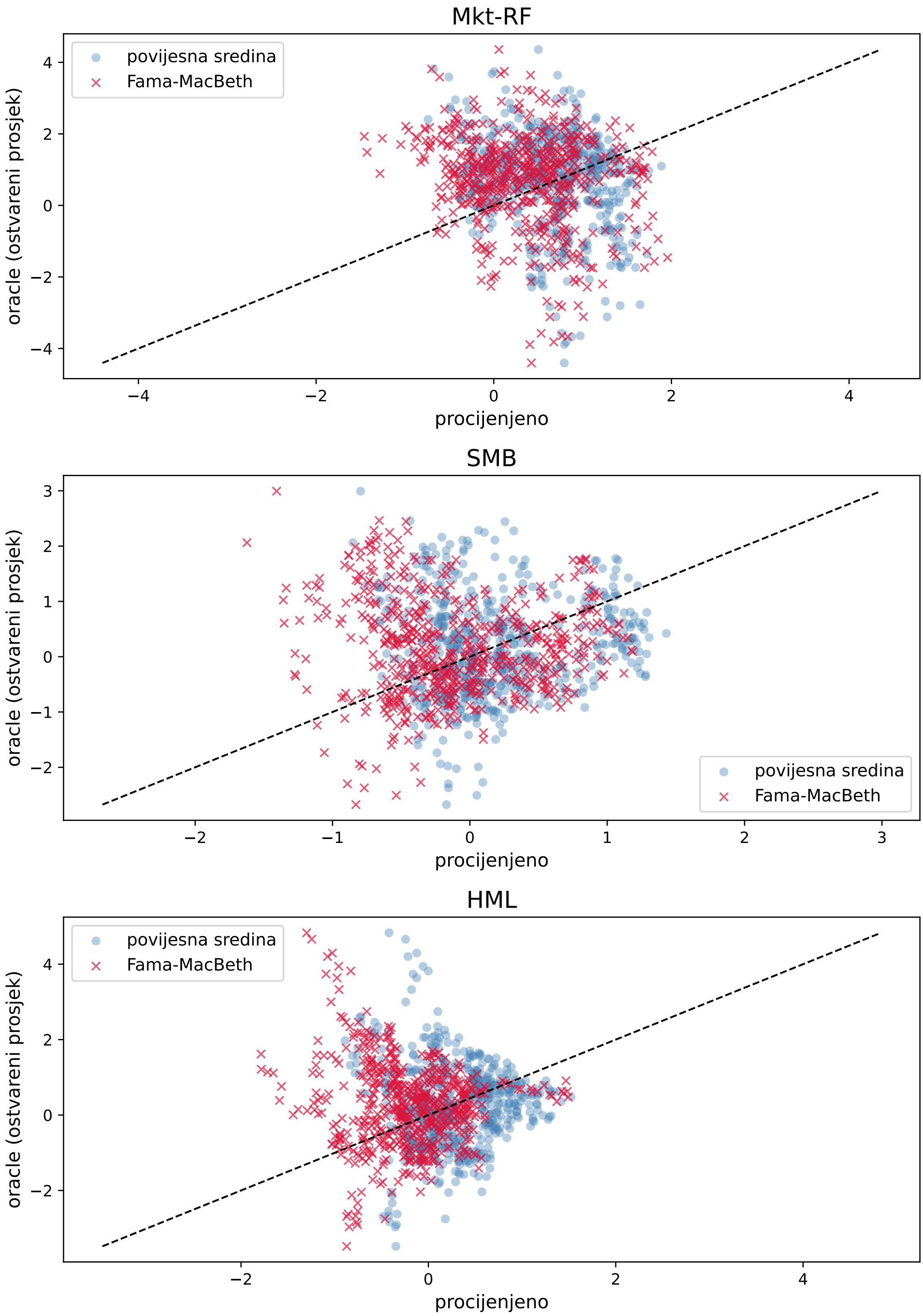

Figure 4.1: Estimates of the factor premiums by the historical mean and by the Fama-MacBeth regression against the oracle value, separately for each factor. Each window position gives one point for each method; the dashed line marks perfect agreement between the estimate and the oracle value.

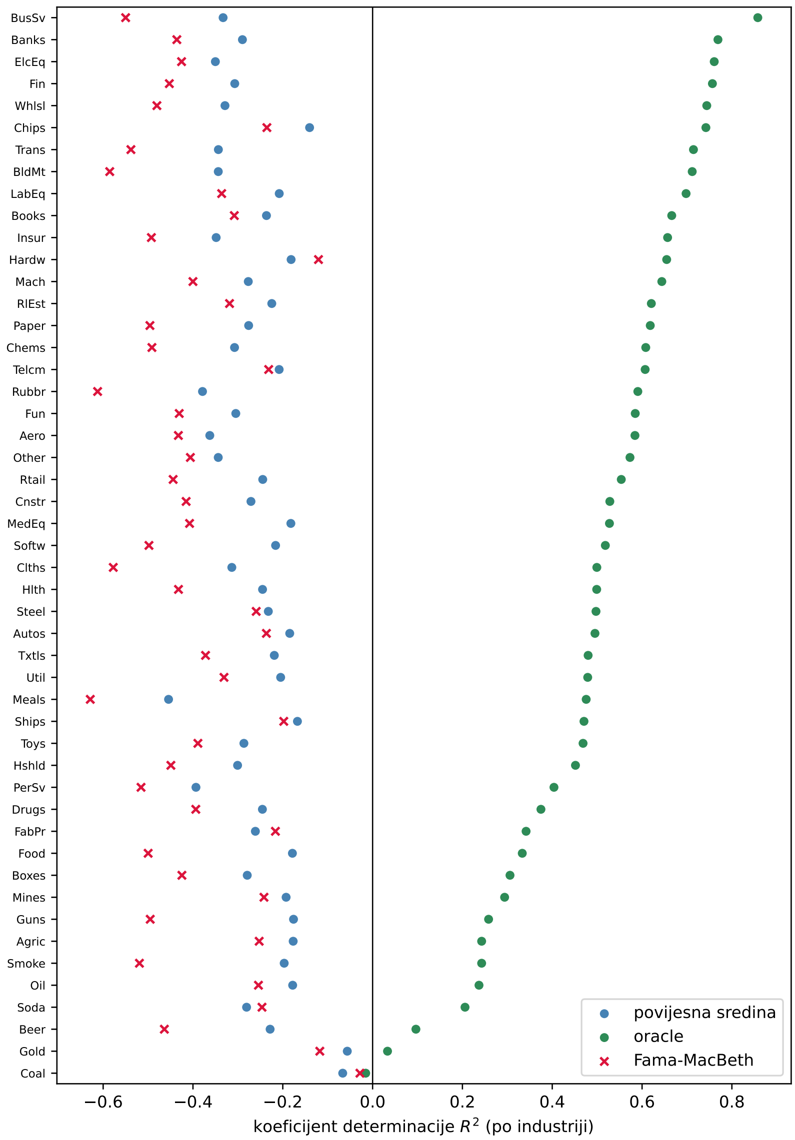

Figure 4.2: Coefficient of determination R^2 for each of the 49 industries for the three classical methods.

and temporarily switches off part of the units and thereby reduces the tendency to overfit, and finally a fully connected layer that gives the three premiums.

We chose such a simple network after trying more complex architectures as well, with more layers and with more units per layer. It turned out that simpler models learn better: the more complex the network was, the more overfitting became a problem, so that such a model would describe the training data well but generalize more poorly to new data. For this reason we settled on a small network with a single LSTM layer.

We train the network by gradually varying its parameters so that the predicted returns match the realized ones as well as possible. The measure that we seek to reduce in doing so is called the loss. In this case it is the mean squared error between the predicted and the realized return, the same quantity as in Equation (4.2). We adjust the parameters in steps that we call *epochs*: in each epoch the network passes once through all the training data and slightly modifies the parameters, using the Adam optimization procedure with a small learning rate, the data divided into batches of 16 examples, and at most 500 epochs.

We divide the windows chronologically into three parts: the first 75 % serve for *training* (451 windows), the next 15 % for *validation* (90 windows), and the remaining part for *testing* (61 windows, starting from 2020). The division is chronological rather than random, so that the network never trains on a period later than the one on which it is subsequently tested. The loss computed on the training set we call the *training loss*, and the one on the validation set the *validation loss*: the first shows how closely the network has fitted the data it has seen, and the second how well it generalizes to data that it did not use in training.

We monitor training across epochs and apply *early stopping*: as soon as the validation loss stops falling and does not improve over a certain number of consecutive epochs, we stop training and restore the parameters from the epoch with the smallest validation loss. In this way we prevent excessively long training from fitting the network too closely to the training data at the expense of generalization. Figure 4.3 shows both losses across epochs. The validation loss falls sharply in the first epochs and then stabilizes. Early stopping selects the best epoch (here epoch 192).

We apply the network trained in this way to the test set, and below we compare its predictions with the classical methods.

4.5 Comparison of the methods and discussion

In order to compare all the methods on the same data, we evaluate them on the test set (the last 61 windows, starting from 2020), which is the only set for which predictions of the LSTM model exist. Table 4.2 shows the R^2 and the MSE of all four methods.

Among the three feasible methods the deep learning model proves to be the best: it has the highest R^2 and the smallest MSE, ahead of the historical mean and apprecia-

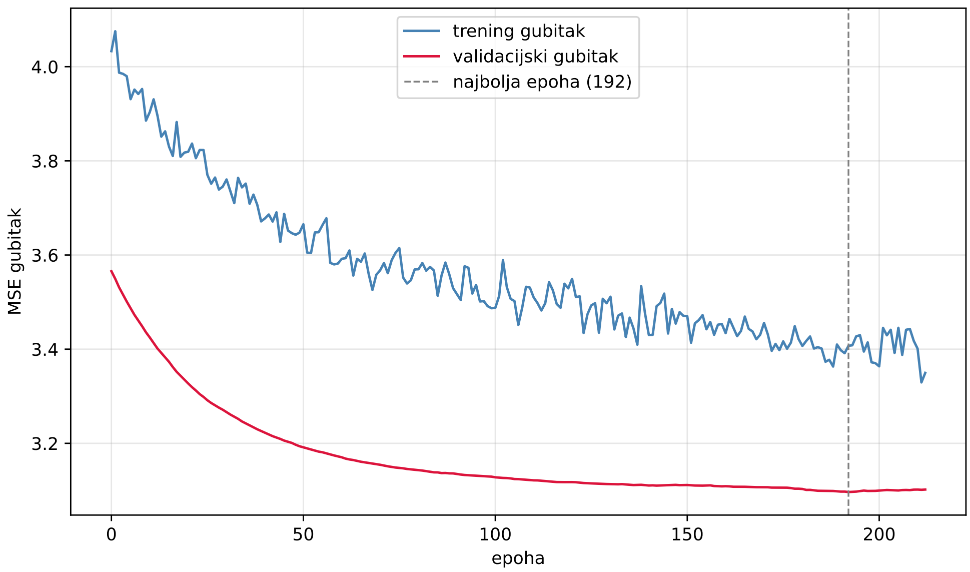

Figure 4.3: Training and validation loss (mean squared error) across the training epochs of the LSTM network. The vertical dashed line marks the epoch with the smallest validation loss, which was determined by early stopping.

Table 4.2: Overall coefficient of determination R^2 and MSE on the test set (61 windows from the beginning of 2020, 2989 predictions) for all four methods.

Method	R^2	MSE
historical mean	−0.348	5.248
oracle	0.380	2.415
Fama-MacBeth	−0.579	6.146
LSTM	−0.141	4.442

bly ahead of the Fama-MacBeth regression, which over this period is the weakest. The network therefore manages to predict the premiums better than both feasible classical methods. Nevertheless, its R^2 is still negative, so it too does not surpass the simple null model, and it remains considerably distant from the idealized upper bound set by the oracle.

Figure 4.4 compares the estimates of the historical mean and of the LSTM model with the oracle value, by factor. With both methods the estimates are widely scattered around the line of equality, so the advantage of the LSTM model lies not so much in a visual match with the oracle value as in a smaller overall prediction error.

To show that the advantage of the LSTM model did not arise merely from aggregation across industries, Figure 4.5 shows the MSE for each individual industry. For most industries the LSTM model has a smaller error than the historical mean and the Fama-

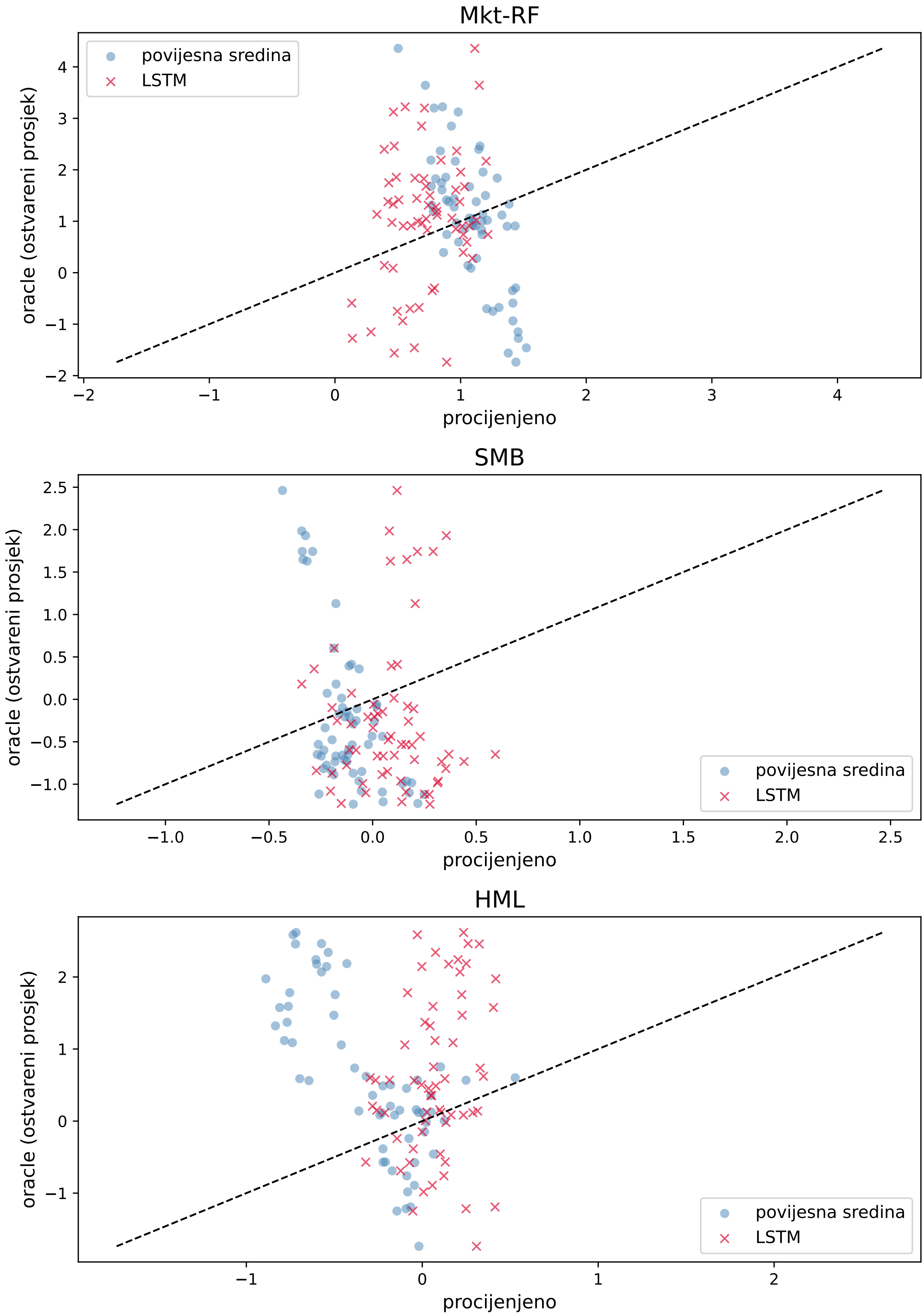

Figure 4.4: Estimates of the factor premiums by the historical mean and by the LSTM model against the oracle value, separately for each factor, on the test set. The dashed line marks perfect agreement between the estimate and the oracle value.

MacBeth regression, while the oracle, as expected, has the smallest; the pattern is very clear across the entire cross-section of industries.

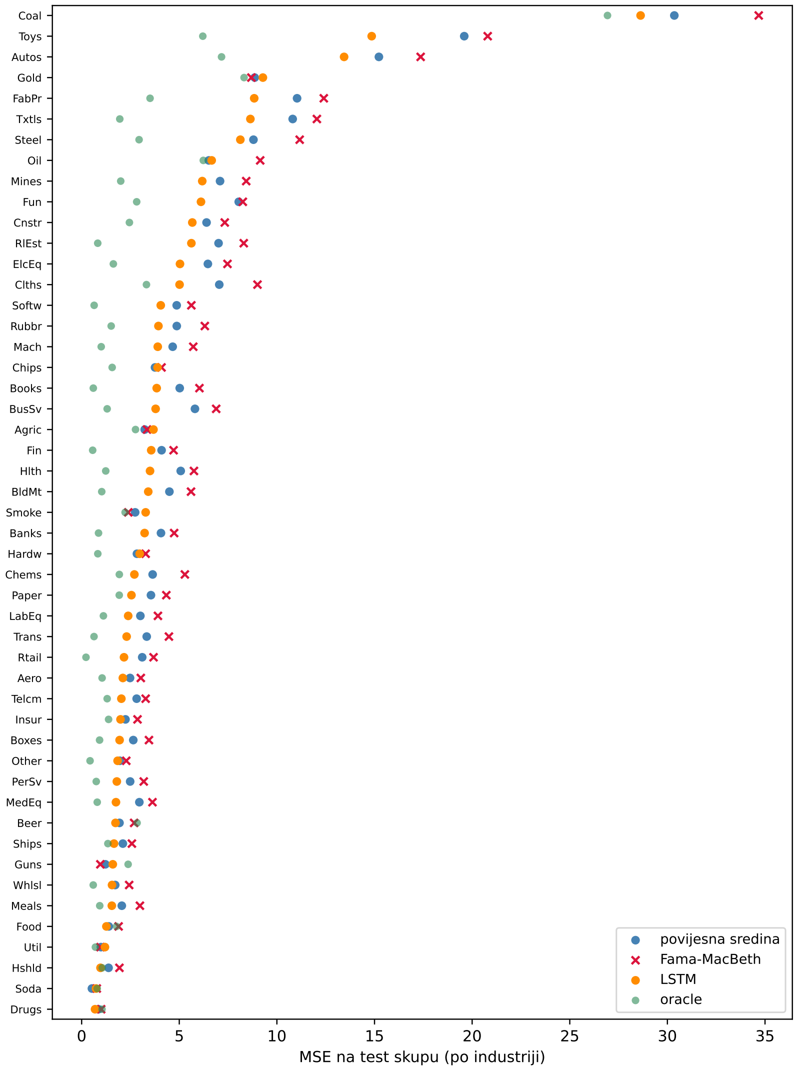

Figure 4.5: Mean squared error on the test set for each of the 49 industries for all four methods, sorted by the value for the LSTM model.

The results show that deep learning brings a consistent, though moderate, improvement over the classical feasible methods for estimating the premiums. Such results support the initial claim of the thesis that machine learning methods can make better use of the patterns in the history of the factors than a simple average. At the same time, the fact that no feasible method achieves a positive R^2, as well as the large gap to the oracle value, confirms how difficult a problem the prediction of the premiums is: financial data are extremely noisy, the premiums change over time, and the test period, which covers the turbulent years from 2020 onwards, makes prediction even harder. The large gap to the oracle bound also shows that the data still contain a considerable amount of predictable information that none of the methods considered manages to exploit fully.

5 Conclusion

The estimation of the risk premiums is the most sensitive step in the application of factor models: how well the model predicts future returns depends largely on it, and the classical estimates are often unreliable. This thesis examined whether a machine learning method can give a better estimate of the premiums than classical approaches. Taking the Fama-French three-factor model and 49 industry portfolios as test assets, we evaluated four methods for estimating the premiums by an out-of-sample comparison with a rolling window: estimation by the historical mean, the Fama-MacBeth regression, the idealized oracle estimate as an upper bound, and estimation by an LSTM network, whose input, alongside the history of the factors, also includes a macroeconomic variable.

The results show that both feasible classical methods (the historical mean and the Fama-MacBeth regression) do not manage to predict the premiums better than the simple average of each industry's returns, so that their coefficient of determination is negative. The estimate based on the LSTM network surpasses both methods, which confirms that machine learning can make better use of the patterns in the history of the factors than a simple average. Nevertheless, it too does not achieve a positive coefficient of determination, nor does it come close to the idealized oracle estimate.

The improvement that deep learning brings is therefore real, but moderate. The large gap between all feasible methods and the oracle bound shows that the data still contain a considerable amount of predictable information that none of the methods manages to exploit fully. This confirms that the estimation of the premiums remains a complex problem: financial data are extremely noisy, and the premiums change over time and are hard to predict reliably.

The results obtained should be viewed subject to several limitations. The test period is relatively short and covers the turbulent years from 2020 onwards, the model used is simple, and only a single macroeconomic variable was taken as an additional input. Further work could move in the direction of richer input data, different network architectures, or a longer and more varied test period, which would establish more precisely to what extent machine learning can improve the estimation of risk premiums.

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Sažetak

Procjena faktora u faktorskim modelima povrata imovina zasnovana na strojnom učenju

Damjan Crnković

Faktorski modeli objašnjavaju povrate imovine njihovom izloženošću zajedničkim izvorima rizika, pri čemu svaki faktor nosi premiju, odnosno prosječnu nagradu za preuzeti rizik. Procjena tih premija najteži je dio primjene modela jer su procjene iz povijesnih podataka često nepouzdane. U radu se ispituje mogu li metode strojnog učenja procijeniti premije bolje od klasičnih pristupa. Na povratima 49 industrijskih portfelja uspoređuju se procjena povijesnim prosjekom, Fama-MacBeth regresija i povratna ćelija s dugoročnom memorijom, uz idealiziranu procjenu koja služi kao gornja granica uspješnosti. Rezultati pokazuju da neuronska mreža nadmašuje klasične primjenjive metode, no procjena premija i dalje ostaje težak i zahtjevan zadatak.

Ključne riječi: faktorski modeli; premija na rizik; procjena premija; strojno učenje; LSTM mreža; Fama-French model; industrijski portfelji

Abstract

Machine learning based factor estimation in factor models of asset returns

Damjan Crnković

Factor models explain asset returns through their exposure to common sources of risk, where each factor carries a premium, that is, the average reward for bearing that risk. Estimating these premiums is the hardest part of applying the model, because estimates from historical data are often unreliable. This thesis examines whether machine learning methods can estimate the premiums better than classical approaches. Using the returns of 49 industry portfolios, it compares estimation by the historical mean, the Fama-MacBeth regression, and a neural network with long short-term memory, together with an idealized estimate that serves as an upper bound on performance. The results show that the neural network outperforms the feasible classical methods, yet estimating the premiums remains a hard and demanding task.

Keywords: factor models; risk premium; premium estimation; machine learning; LSTM network; Fama-French model; industry portfolios